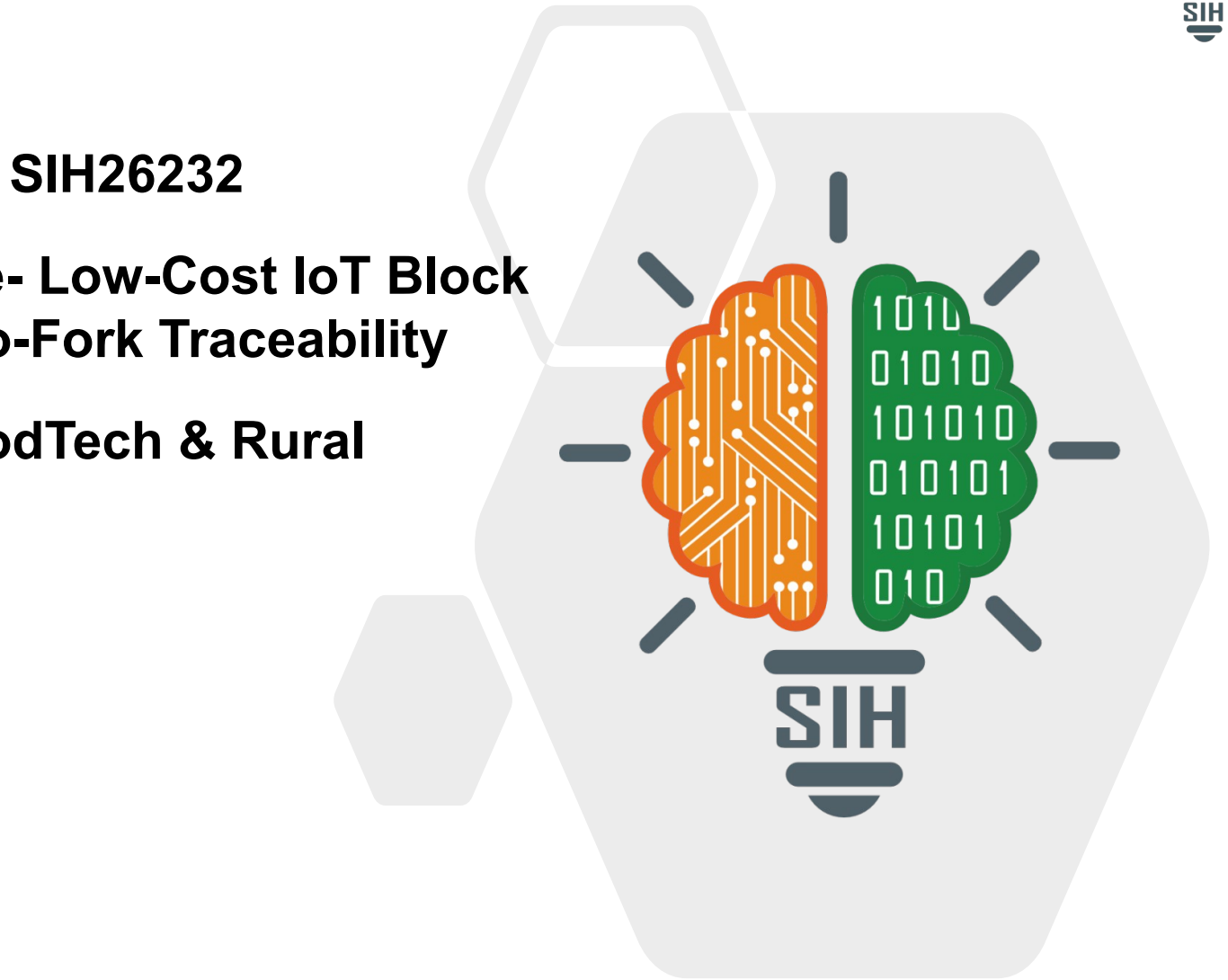
SMART INDIA HACKATHON 202

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- Problem Statement ID – SIH26232
- Problem Statement Title- Low-Cost IoT Block chain Nodes for Farm-to-Fork Traceability
- Theme- Agriculture, FoodTech & Rural Development
- PS Category- Hardware
- Team ID- 159400
- Team Name: INVICTUS



INVICTUS

SmartCool: Solar-Powered Smart Mini Cold Storage

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Proposed Solution

A decentralized, solar-powered mini cold-storage unit for farm-gate, village collection centres and FPOs : a temporary storage buffer between harvest and transport.

Insulated chamber + DC compressor refrigeration + forced-air circulation, fed by Solar PV → MPPT → Battery → DC bus for unreliable-grid and off-grid sites.

ESP32 closed-loop control with hysteresis holds crop-specific temperature and humidity; energy-aware modes (Normal / Energy-Aware / Critical) protect produce on low battery.

Fault detection for high temperature, door-open, low battery, solar-charging and cooling failure, with local buzzer / LED / display alerts that need no internet.

Offline-first: RTC + MicroSD logging with store-and-forward sync (Wi-Fi / GSM) and a simple crop-select interface; set-points validated on the prototype.

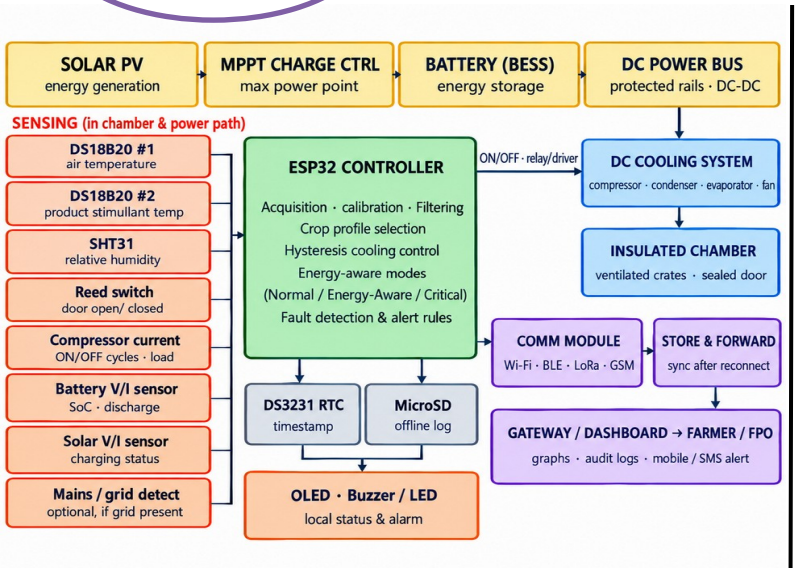
Blockchain traceability: every sensor record is SHA-256 hashed and anchored to a blockchain ledger, giving a tamper-proof farm-to-fork audit trail.

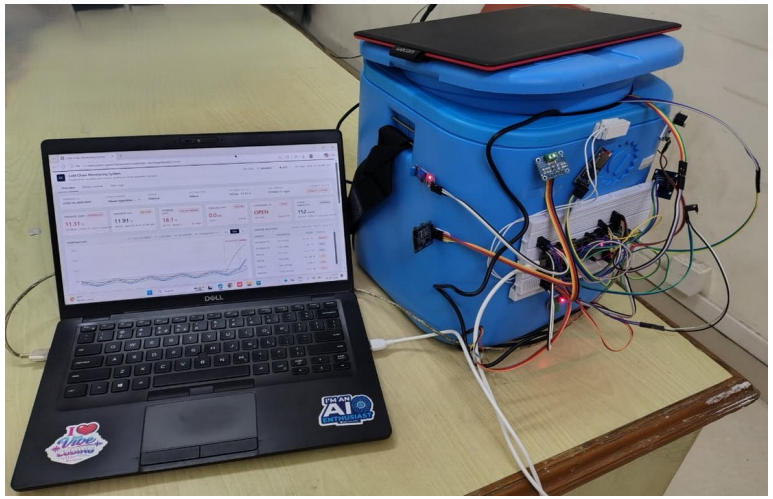


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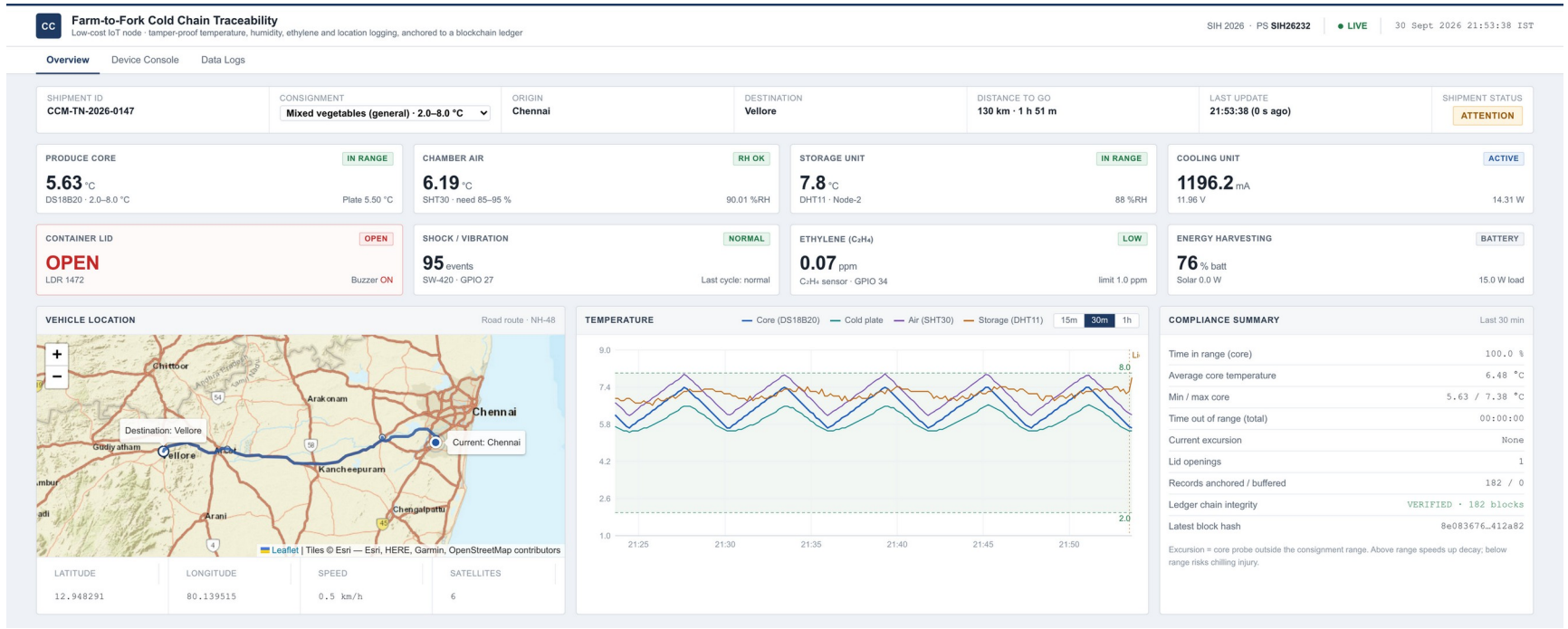
TECHNICAL APPROACH

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Prototype



Dashboard

Techstack Used:

1. Solar PV + MPPT controller
2. Battery (BESS) + DC protection
3. DC compressor refrigeration
4. PU/PIR insulated chamber
5. ESP32 (Arduino / ESP-IDF C++)
6. Digital temp + RH sensors
7. Reed switch, V/I sensors
8. RTC + MicroSD, OLED/LCD
9. Wi-Fi / GSM / LoRa (optional)
10. Web dashboard (optional)

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FEASIBILITY AND VIABILITY

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Analysis of the feasibility of the idea	Potential challenges and risks	Strategies for overcoming these challenges
Technical – Achievable	**Sizing solar, battery and cooling load for NER climate**	Cooling-load and energy calculations before hardware selection; proven DC refrigeration, MPPT and standard sensors; ratings validated on the prototype.
Reliability – Resilient	**Cloudy / rainy days, night, grid outages, no network**	Solar + battery with energy-aware / critical modes; offline-first control, alerts and logging; fault detection for cooling, door and solar failure.
Operational – Practical	**Unreliable power & connectivity; limited cold-storage access; limited technical support**	Local data storage + cloud synchronization, Safe Holdover Time Prediction, intelligent alerts, farmer-friendly crop selection, local status display, offline-first operation, weather-resistant modular unit.

Given that feasibility and viability are our top priority we will ensure to keep everything

- **Technically viable –** uses existing solar, DC refrigeration and ESP32 technology; one or two crops validated first.
- **Economically feasible –** small modular units for FPO / collection-centre deployment; cost to be worked out after design.
- **Environmentally sound –** renewable solar energy, energy-efficient cooling, designed to cut post-harvest and transport losses.

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IMPACT AND BENEFITS

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Potential Impact on Target Audience

1. Farmers
- Reduces forced immediate sale and spoilage; flexible transport timing.

2. FPOs & Cooperatives
- Shared, monitored local storage at aggregation points.

3. Village collection centres / traders
- Better produce handling before transport to markets.

4. Remote NER communities
- Cold-chain access without depending on grid supply.

5. Government / MDoNER
- Scalable decentralized horticulture cold-chain model.

Benefits of the solution

1. Social
- Improves farmer bargaining flexibility; supports cooperatives.

2. Economic
- Improves income opportunity by cutting post-harvest losses.

3. Environmental
- Renewable solar energy; less wasted produce and transport.

4. Technological
- Energy-aware, offline-first rural embedded system.

5. Strategic
- Replicable modular units across horticultural clusters.

Validated on the working prototype:

Temperature stability	Cooling response	Solar charging	Battery backup	Safe Holdover Time Prediction	Alert response	Offline log retention	Vegetable quality check

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RESEARCH AND REFERENCES

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Research Survey

Aspect	Conventional Systems	Proposed System
Location	Large centralized cold rooms, far from farms	Decentralized farm-gate / village mini units
Power	Grid-dependent	Solar + battery, energy-aware operation
Control	Basic thermostat	Closed-loop, crop-specific profiles
Monitoring	Manual checks, network-dependent	Offline-first logging + local alerts
Faults	Found after damage	Cooling / door / solar fault detection
Traceability	Paper records, easily altered	Blockchain-anchored, tamper-proof sensor logs

Reference

[1] FAO, “Global Food Losses and Food Waste – Extent, Causes and Prevention”, Rome (2011).

[2] ICAR-CIPHET, “Assessment of Quantitative Harvest and Post-Harvest Losses of Major Crops and Commodities in India”, Ludhiana (2015).

[3] National Centre for Cold-chain Development (NCCD) and National Horticulture Board (NHB): cold-chain infrastructure and horticulture data.

[4] MNRE / NABARD: solar-powered cold storage and agri-infrastructure guidelines.

[5] Peer-reviewed literature on solar PV–battery DC refrigeration and small-scale vegetable cold storage (to be shortlisted).

GitHub Repository (code, firmware, dashboard)
github.com/Sanmugavel1/SIH2026-FarmToFork-ColdChain

Solution Video
youtu.be/o2jbTQCLLzo